



Review article

Placenta accreta spectrum disorder: updates on anesthetic and surgical management strategies

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ABSTRACT

Placenta accreta spectrum (PAS) is a leading contributor to major obstetric hemorrhage and severe maternal morbidity in the developed world. In the United States, PAS has become the most common cause of peripartum hysterectomy. Over the last 40 years, clinicians have also witnessed a dramatic increase in the incidence of PAS. In the 1950s, the incidence of PAS was reported to be 0.03 per 1000 pregnancies. Recent epidemiological studies estimate that the PAS incidence is between 0.79 and 3.11 in 1000 pregnancies. As a consequence, obstetric anesthesiologists are increasingly likely to be called upon to manage women with suspected PAS for delivery. Given the increasing incidence and the morbidity burden associated with PAS, anesthesiologists play a vital role in optimizing maternal outcomes for women with PAS. This review will provide up-to-date information on nomenclature, pathophysiology, risk factors, antenatal detection, systemic preparations (includes timing of delivery, location of surgery, pre-operative evaluation and patient positioning), surgical and anesthetic approach, intra-operative management, invasive radiology and postoperative plans.

Introduction

Placenta accreta spectrum (PAS) is a leading contributor to major obstetric hemorrhage and severe maternal morbidity in the developed world.¹ In the United States (US), PAS has become the most common cause of peripartum hysterectomy.² While in the 1950s the incidence was 0.03 per 1000 pregnancies,³ based on recent epidemiological data the incidence is between 0.79 and 3.11 per 1000 pregnancies.⁴ This review of PAS provides up-to-date information for obstetric anesthesiologists on the nomenclature, pathophysiology, epidemiology, clinical diagnosis and peri-operative management of patients with PAS. Throughout the review, we describe key aspects of the authors' practices that are relevant to the anesthetic management of patients with PAS.

Nomenclature

The term PAS refers to a broad spectrum of disease from pathological adherence to widespread extension of the placenta into the uterus and other surrounding viscera. Other terms like abnormally invasive placenta and morbidly adherent placenta are also used, but are not favored. Placenta accreta spectrum is traditionally classified into three subtypes (accreta or creta, increta, and percreta). Although readers are

likely familiar with this classification scheme, newer clinical and pathological grading schemes focused more heavily on the exact location of involvement have been proposed by the International Federation of Gynecology and Obstetrics (FIGO) and other organizations.^{5,6}

Pathophysiology of placenta accreta spectrum

In a patient with severe PAS, normal uterine structure is dramatically altered and the maternal pelvis is transformed into a superhighway of hypervascularity that can result in massive hemorrhage at delivery. Although incompletely understood, the pathophysiology of PAS is most likely due to the presence of defective uterine decidua, abnormal attachment of the trophoblast directly to the myometrium, and pathologic transformation of both the placenta and uterus.^{5,6} Defective uterine decidua is the result of instrumentation and scarring from prior uterine surgery, such as cesarean delivery. During implantation, trophoblasts encounter a uterine scar instead of uterine decidua. In this foreign environment, the trophoblast is dysregulated. The typically ordered interaction of the early placenta and maternal tissues is disrupted,⁷ resulting in abnormal attachment, uterine scar dehiscence, extension of the placenta into or through the wall of the uterus, and insidious transformation of the pelvic vasculature.

Epidemiology and risk factors for placenta accreta spectrum

The most common and important risk factors for PAS are a history of prior cesarean delivery and placenta previa. A 'dose-response' rela-

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tion exists between the risk of PAS and the number of prior cesarean deliveries. The PAS risk is 0.6% and 2.1% in women with a history of two and three prior cesarean deliveries, respectively.^{8,9}

A patient's risk of PAS increases substantially when a placenta previa is present. Patients with a placenta previa have a 35- to 65-fold increased odds of PAS compared with those without a placenta previa.¹⁰⁻¹² In terms of absolute risk, among patients with two prior cesarean deliveries, those with a placenta previa have a 40% absolute risk of PAS; this contrasts with a risk below 1% for those without a placenta previa.⁸ Based on these findings, obstetricians and anesthesiologists should be vigilant for PAS in women with a placenta previa and any number of prior cesarean deliveries, even if no other radiographic signs of accreta are present. Other reported risk factors for PAS include advanced maternal age, in vitro fertilization, low-lying anterior placenta (in the setting of a prior cesarean delivery), and history of focal accreta or retained placenta.¹³

Antenatal detection

Patients managed at an experienced specialist center have a lower risk of adverse maternal morbidity than those managed in other centers.^{14,15} Thus, based on current US and United Kingdom (UK) obstetric society guidelines, patients with PAS clinical risk factors and suggestive ultrasound features should have early referral to a specialist center.^{16,17} Unfortunately, many patients will not deliver in a specialist center because of a missed diagnosis (which occurs in 50% of patients with PAS)^{11,18} and a lack of standardized processes or referral pathways for women with suspected PAS.¹⁹ A US survey published in 2013 reported that fewer than a quarter of obstetricians refer patients with PAS to a specialist center.²⁰

Ultrasound is the primary diagnostic modality for the antenatal diagnosis of PAS.^{16,17,21} An adjunct imaging modality is magnetic resonance imaging (MRI), but its diagnostic utility is less clear. Magnetic resonance imaging may be helpful when ultrasound findings are equivocal or when placental location is not predominantly anterior. However, MRI may not correctly confirm the pre-operative diagnosis or improve clinical management.²² Ultrasound and MRI have high sensitivity and specificity (from 76% to 94% and 74% to 100%, respectively) for detecting PAS, but these estimates should be interpreted with caution.^{23,24} Studies that reported diagnostic accuracy were performed in optimized settings, involved cohorts at very high risk for PAS, and were performed by experienced physicians in high-volume specialist centers. Thus, the reported diagnostic accuracy of both ultrasound and MRI may be inflated. A detailed description of sonographic and MRI markers for PAS is outside the scope of this review but has been described.^{13,25,26} Unfortunately, a reliable and reproducible biomarker for PAS is not yet available;²⁷ but discovery work is underway.²⁸

Preparation for placenta accreta spectrum

Hospital system preparation in placenta accreta spectrum specialist centers

Experienced multidisciplinary teams based in a center that manages a high PAS volume are indicative of a specialist center. At these centers, care is closely co-ordinated across multiple teams. The minimum criteria for a PAS specialist center, as suggested by the International Society for Placenta Accreta Spectrum (IS-PAS), are listed in Table 1. Several aspects of enhanced systemic preparedness should be considered at PAS specialist centers. Regular interdisciplinary communication is an important component of delivery planning. Much like 'tumor board' in cancer centers, providers discuss future PAS cases scheduled for delivery at their center. Past cases and pathology are reviewed to discover opportunities for improvement in diagnosis, management, interdisciplinary consensus-building and co-ordination.

Table 1

Criteria for a placenta accreta spectrum specialty center

1. A multidisciplinary team available 24 h a day, 7 days a week that includes:
 - a. Imaging expertise
 - b. Experienced obstetrician (often maternal-fetal medicine)
 - c. Anesthesiologist with complex obstetrical expertise
 - d. Surgeon experienced with complex pelvic surgery (often gynecologic oncology)
 - e. Urologist
 - f. Neonatologist
 - g. Interventional radiologist
2. Access to:
 - a. Colorectal surgeon
 - b. Vascular surgeon
 - c. Hematologist or blood bank specialist
3. Adult intensive care facilities
4. Neonatal intensive care facilities
5. Massive transfusion facilities
6. Intra-operative blood salvage services

Adapted from Collins et al. IS-AIP guideline for management of abnormally invasive placenta. *Am J Obstet Gynecol* 2019.³⁵

A specialist team should ideally comprise physicians from the following disciplines: maternal-fetal medicine, gynecology-oncology, obstetric anesthesiology, general and trauma surgery, interventional radiology (IR) and neonatology. Because up to 50% of patients with PAS undergo unscheduled preterm delivery before 34 weeks' gestation,²⁹ anesthesia and surgical teams need to be mobilized at short notice in the event of an urgent or emergency cesarean hysterectomy. Therefore, specialist centers should have sufficient staffing availability to cover all hours.

Key resources

Critical hospital resources include a blood bank that can support massive transfusion, an intensive care unit with staff experienced in managing critically ill postsurgical patients, and the capability to manage preterm babies. Centers that lack these resources require clear indications and guidelines for referral to a PAS specialist center.¹⁹ Table 2 presents potential indications for referral. Development of hospital-specific protocols for pre-, peri- and postoperative management may reduce error in a complex system, enhance interdisciplinary trust, and improve quality of care over time.³⁰ For example, specialist centers should have checklists, team learning, and simulation.³¹⁻³³

Timing of delivery

Decisions related to the timing of delivery are fraught with competing maternal and neonatal risks. Scheduled delivery, earlier in gestation, reduces the maternal risk associated with an emergency

Table 2

Consider referral to centers of excellence for diagnosis and management in patients with clinical or sonographic risk factors for placenta accreta spectrum.

- | |
|--|
| Clinical risk |
| 1. Multiple prior cesarean sections |
| 2. Placenta previa |
| 3. History of endometrial ablation |
| 4. Previous uterine surgery (e.g. operative hysteroscopy, myomectomy, septum resection) |
| 5. First- or second-trimester bleeding with other risk factors for placenta accreta spectrum |
| 6. In vitro fertilization |
| Sonographic risk |
| 1. Abnormal placental appearance |
| 2. Abnormal uterine shape |
| 3. Abnormal vascularity of myometrial wall |
| 4. Cesarean scar pregnancy |

Adapted from Silver et al. Center of excellence for placenta accreta. *Am J Obstet Gynecol* 2015.¹⁹

cesarean delivery, but increases the neonatal risks associated with prematurity. In contrast, delivery later in gestation increases the risk of an emergent delivery. Based on recommendations from the American Society of Obstetricians and Gynecologists (ACOG) and the Society for Maternal-Fetal Medicine,³⁴ the Royal College of Obstetricians and Gynaecologists (RCOG),¹⁷ and IS-PAS,³⁵ delivery should not occur before 34 weeks' gestation if a patient has no bleeding, rupture of membranes, or preterm labor. These guidelines are based on low-quality evidence, so the timing of delivery should be planned according to risk factors for unplanned delivery, and whether a center can mobilize a team for an urgent or emergency cesarean hysterectomy.

Location of surgery

Limited data exist to determine the ideal surgical location for delivery. Options include a regular or hybrid (radiology equipment) operating room in the main adult hospital, in the labor and delivery unit, or in the IR department.³⁶ In a US survey of 84 obstetric anesthesia division chiefs,³⁷ 29% of respondents reported that the labor and delivery unit was the preferred surgical location for higher risk cases (such as increta or percreta). A higher proportion (79%) of respondents used the labor and delivery unit for 'low-risk' cases.³⁷ Based on providers' preferences, hospital layout, clinical workflows, access to key resources and blood bank, the optimal location for PAS surgery should be a local decision.

Anesthesia pre-operative evaluation

The anesthesiologist should be involved early in delivery planning for PAS cases.^{19,38} Early involvement allows sufficient lead-time for a full anesthetic assessment, identification of unique patient needs, and medical optimization from subspecialists such as cardiology, endocrinology, and sleep medicine. Physicians can apply patient-blood management practices by ensuring that women with antepartum anemia undergo diagnostic testing and treatment to optimize their hemoglobin level before delivery. A pre-anesthesia checklist is a useful tool for describing key information for peri-operative planning and communication, for example equipment needs, names and contact details of key personnel, blood product ordering processes, and plans for postoperative disposition. Several examples of these checklists are available.^{31,33}

Intra-operative management

Despite the importance of radiological imaging to identify women with suspected PAS, it is often diagnosed clinically by the surgical team at delivery or by a pathological examination of the resected uterus. Key aspects of the intra-operative management at each author's institution are summarized in Table 3.

Interventional radiology

Invasive radiology interventions (occlusion or embolization) have been described for the peri-operative management of PAS patients. To reduce intra-operative bleeding, placement of arterial balloons before surgery in the abdominal aorta, common iliac, internal iliac or uterine arteries has been reported.³⁹ During surgery, the balloons can be inflated to reduce arterial blood flow to the uterus. Embolization is a component of the postpartum hemorrhage protocol in some centers, for women who experience postoperative bleeding. The patient should be stable enough for transport to the angiography suite.

Current evidence to support IR interventions (vascular balloon occlusion, embolization) is of low quality, comprising mainly case series, observational and retrospective studies, and to date only three randomized controlled trials (RCTs) have been published.^{40–42} All three RCTs reported no difference in estimated blood loss, when patients

Table 3

Peri-operative and postoperative approaches for the management of women with placenta accreta spectrum at the University of Utah and Tel-Aviv Sourasky Medical Center.

	University of Utah, US	Tel-Aviv Sourasky Medical Center, Israel
Pre-operative multidisciplinary communication	Encrypted email and via electronic medical record routing	Via smartphone group messaging
Pre-operative multidisciplinary meeting	Yes, monthly treatment planning conferences	No
Pre-operative debrief	Yes	Yes
Place of surgery	Main operating room for all suspected cases of PASL & D operating rooms for cases with very low suspicion	Scheduled: hybrid operating room in main operating room Emergent/urgent: L & D operating rooms.
Invasive radiology	Not routine. Post-delivery embolization on a rare case-by-case basis	Scheduled cases, internal iliac balloon placement, inflated after delivery, prior to placental manipulation Emergent/urgent: not used
Primary anesthesia mode	Scheduled: Neuraxial (CSE) for cesarean with intra-operative conversion to general anesthesia after delivery Emergent/urgent: General anesthesia	Scheduled: Neuraxial (CSE) Emergent/urgent: General anesthesia
Intravenous lines	2 × 14-gauge peripheral cannulae	2 × 14-gauge peripheral cannulae
Rapid infuser device	Yes	Yes
Cell salvage	Not routine	Not routine
Arterial line placement	Yes	Yes
Central line placement	Not routine	Not routine
Pre-operative blood ordering before scheduled cesarean hysterectomy	6 units PRBC; 6 units plasma	8 units PRBC (4 units available in the operating room)
Point-of-care devices	Arterial blood gas machine and thromboelastometry	Arterial blood gas machine in operating room Thromboelastometry (in hematology laboratory)
Atony prophylaxis with oxytocin	Yes, after delivery of the fetus	Yes, after delivery of the fetus
Tranexamic acid	If actively bleeding after delivery of fetus	If actively bleeding
Postoperative disposition	Routine: Postoperative OBGYN care unit (postpartum); Complicated: SICU	Routine: L & D postoperative care unit (staffed by intensive care trained nurses) Complicated: General ICU
Postoperative analgesia	Intrathecal morphine; TAP or QLB block; oral acetaminophen and NSAIDs; oral oxycodone or IV hydromorphone for breakthrough pain	Intrathecal morphine; TAP or QLB block if no neuraxial block PACU: intravenous acetaminophen, metamizole and ketorolac Department: oral acetaminophen, metamizole, NSAIDs, and tramadol for breakthrough pain

Key: L & D: labor and delivery; CSE: combined spinal-epidural anesthesia; IV: intravenous; PRBC: packed red blood cells; SICU: surgical intensive care unit; ICU: intensive care unit; OBGYN: obstetrics and gynecology; TAP: transversus abdominis plain block; QLB: quadratus lumborum block; NSAIDs: non-steroidal anti-inflammatory drugs; PACU: postoperative anesthesia care unit

did or did not undergo an IR procedure for their delivery. Studies using IR for PAS cases are summarized in two systematic reviews.^{39,43} The search strategies led to identification of different studies, and only eight overlapped, including one RCT.⁴¹ D'Antonio et al. reviewed 15 studies involving 958 cases, including only studies that compared IR

with no IR,⁴³ after an antenatal diagnosis of PAS. Shahin et al. reviewed 69 reports, and performed a meta-analysis on 14 studies that compared IR with no IR.³⁹ Both reviews reported significantly decreased blood loss when IR was used: D'Antonio et al. a mean difference of -1.0 L (95% confidence intervals (CI) -1.6 to -0.4 L),⁴³ and Shahin et al. a mean difference of -0.9 L (95% CI -1.4 to -0.4 L). Complications reported include limb or buttock claudication, arterial thrombosis, balloon rupture or migration, and uterine necrosis (requiring hysterectomy).³⁹ Blood loss was lowest using abdominal aortic occlusion, according to four studies, and in these four studies there was a 73-fold reduction in the odds of hysterectomy.

Patient positioning

To optimize surgical exposure, patients should be in the lithotomy position to allow access to the perineum for a pre-delivery cystoscopy and cervix for the final steps of hysterectomy. However, the lithotomy position is not preferred if balloon catheters have been placed pre-operatively. To minimize the chance of angio-balloon displacement, the urinary catheter and neuraxial block should be placed first. Regardless of patient positioning, the surgical and anesthesia teams should regularly evaluate the extent of blood loss between the patient's legs, as this may be concealed but nevertheless account for a large proportion of the overall blood loss.

Surgical approach

There is no single standard treatment for PAS and surgical practices vary significantly worldwide.³⁵ Surgical approaches are classified as radical or conservative. Hysterectomy is the cornerstone of radical treatment. Uterine-sparing approaches characterize conservative approaches.

Radical surgery (hysterectomy)

The term radical relates to the extent of the surgical procedure and the perceived need to remove the entire uterus and all involved tissue. The majority of PAS centers in the US recommend planned hysterectomy at the time of delivery (immediate cesarean hysterectomy) as the standard of care.⁴⁴ A recent survey of international PAS experts found that 61% preferentially perform immediate hysterectomy.⁴⁵ Another radical approach is delayed or interval hysterectomy. In this strategy, the surgical team performs an intra-operative evaluation of PAS severity and, if immediate hysterectomy is expected to be difficult because of advanced disease (Fig. 1), the decision is made to delay hysterectomy.^{46,47} After the cesarean birth, the uterus is closed with the placenta in situ for 4–6 weeks to allow partial placental resorption, uterine involution, and regression of pelvic hypervascularity. After this period, a scheduled hysterectomy can be performed. Proposed benefits of delayed hysterectomy are decreased intra-operative blood loss and a lower risk of urinary tract injury. Conversely, there is the risk of an unplanned hysterectomy for massive unpredictable hemorrhage and disseminated intravascular coagulation (DIC) during the period between delivery and hysterectomy. After delivery, the placenta is not sterile and septic shock, uterine necrosis, and maternal death have been reported in women awaiting an interval hysterectomy.^{46,48} The optimal approach and timing for hysterectomy in PAS care is not yet known.^{49,50}

Conservative surgery (uterine-sparing)

Conservative management strategies focus on the avoidance of a hysterectomy. When suspicion for PAS is very low, and significant pelvic hypervascularity is not present, a trial of placental removal may be attempted.⁵¹ Under these circumstances, the team should be prepared for a hysterectomy in the event of rapid hemorrhage following placental separation. Other variations on placental removal have been reported, including partial resection of the involved uterine wall and

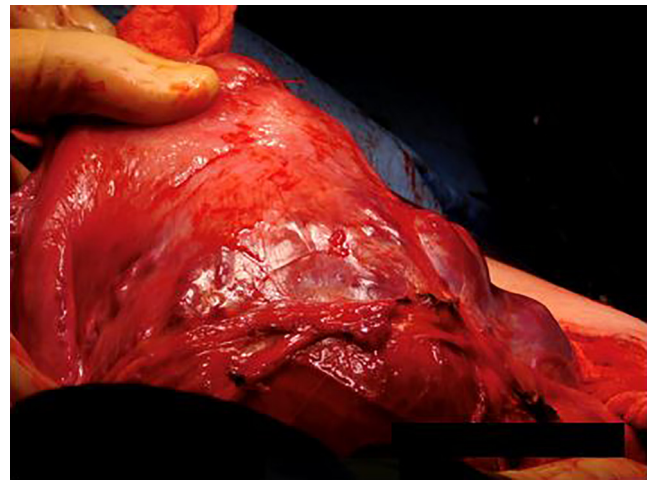


Fig. 1. Intra-operative photograph of the uterus, exteriorized through a midline vertical skin incision after delivery of the neonate. Normal uterine fundus is held in the operator's hand (left), and abnormal placental involvement and vascular transformation of the pelvis is apparent (right). This represents higher grade placenta accreta spectrum (FIGO grade 3).

reconstructive uterine surgery at the time of delivery.^{52,53} Cases with limited involvement (focal accreta) may be managed by simple surgical resection or by leaving a portion of the placenta in situ. In Israel, due to desired high fecundity, uterine-sparing surgery is often favored, and a step-wise surgical approach may enable the decision regarding placental removal or hysterectomy to be taken during surgery.⁵⁴

The last common conservative strategy is in-situ or expectant management. The placenta is retained within the uterus until full placental resorption occurs. In France, this approach has become the standard of care. In a cohort study of 167 patients, severe maternal morbidity occurred in 6% of women who underwent expectant management versus 40–50% of women who received usual management. Further, 78% of women who underwent expectant management did not undergo hysterectomy.⁴⁸ However, there is a risk of emergent re-operation if there is secondary bleeding or an infection from a retained placenta. Patients may also experience bothersome vaginal discharge for months or require hysteroscopic resection of retained placental fragments. These patients require frequent and prolonged follow-up after delivery.

Approach to anesthesia

Options for the mode of anesthesia include neuraxial anesthesia, primary general anesthesia or a combined neuraxial-general anesthetic. In the latter case, general anesthesia can be induced after the neuraxial block, for example as a planned conversion in the period between delivery of the fetus and the onset of the placental management and/or hysterectomy. Unanticipated bleeding or hemodynamic instability are reasons for unplanned conversion to general anesthesia. At our respective institutions, we perform either a catheter-based neuraxial technique or a combination with neuraxial anesthesia converted to general anesthesia after delivery (Table 3). Advantages of neuraxial anesthesia include early bonding and good postoperative analgesia with intrathecal morphine. Because surgery for PAS may be prolonged,⁵⁵ a neuraxial catheter-based technique allows extension of the primary neuraxial anesthetic with epidural top-ups.

General anesthesia as the primary mode (for emergency surgery, expected difficult airway or patient request) has several advantages. For example, it alleviates concern about the potentially detrimental effect of a sympathectomy on maternal hemodynamics during active hemorrhage.⁵⁶ Moreover, it obviates the need to perform intubation

in a hemodynamically unstable patient, especially in the presence of a non-reassuring airway. Data from several surveys suggest that anesthesiologists prefer using general anesthesia as their primary anesthesia mode for patients with the most invasive PAS subtypes.^{37,57} General anesthesia may be beneficial for patients with pre-existing anxiety or psychological disorders who are unlikely to tolerate being awake during prolonged surgery.⁵⁸ Disadvantages of general anesthesia include the need for airway manipulations and exposure of the fetus to general anesthesia before delivery; this exposure may be prolonged if adhesions are encountered by the surgical team. At Tel-Aviv Sourasky Medical Center, patients undergoing primary general anesthesia who express anxiety about being awake for placement of intravenous cannulae and the arterial line may be offered general anesthesia prior to line placement. In a small retrospective study, Shaylor et al. observed similar mean Apgar scores in infants exposed to general anesthesia during line placement and surgery compared with only during surgery.⁵⁹ We acknowledge that other anesthesiologists may wish to avoid this additional anesthetic exposure and, instead, use anxiolysis during line placement.

Published data describing anesthesia modes for cesarean hysterectomies are predominantly sourced from single obstetric centers. These data suggest substantial variability in primary anesthesia modes, and in the conversion rates from neuraxial to general anesthesia. For example, in a series of 40 PAS cases,⁶⁰ Taylor et al. initially used neuraxial anesthesia in 9% of women, with a 45% conversion rate to general anesthesia. However, following a change in surgical technique, combined spinal-epidural anesthesia became the prevailing modality. This change was facilitated by less frequent manual placental removal, a concurrent increased rate of cesarean hysterectomy, decreased intra-operative blood loss (from median of 1.7 L to 1.4 L), and a decrease in blood transfusion (from 35% to 14%). Lilker et al. reported 23 cases with a 29% conversion rate from neuraxial to general anesthesia due to massive hemorrhage; however, over time, neuraxial anesthesia became their preferred technique.⁶¹ In 2002, a PAS protocol was created in Hadassah Ein Kerem, Jerusalem, that advocated general anesthesia,⁶² yet by 2013 the choice of anesthesia modality was discretionary,⁶³ reflecting a practice change facilitated by experience. In a descriptive study of 129 women with PAS who delivered at a US tertiary obstetric center, most women, 122 (95%), who underwent non-emergency PAS cases received neuraxial anesthesia as the primary anesthesia mode,⁶⁴ and the conversion rate to general anesthesia was only 16%, mainly due to pain prior to surgery or resuscitation during major hemorrhage. Only seven women (5%) received general anesthesia as the primary mode, all being of higher risk and all then having a cesarean hysterectomy.

Unexpected placenta accreta spectrum

If a patient with PAS requires urgent or emergency delivery at a non-specialist center, alternatives to immediate hysterectomy should be considered (see surgical section above). A checklist template of recommended actions and resources (according to timing of diagnosis pre-delivery or peri-operatively) for unexpected cases of PAS can be found at the Society for Maternal-Fetal Medicine (SMFM) website.⁶⁵

In cases of unexpected PAS, the goals are to address immediate bleeding risks, to assess the need for and obtain resources to address the surgical complexity and high acuity of PAS cases, and to consider alternative surgical treatments or maternal transport. Low acuity centers may be less well equipped to manage these unexpected cases, and a surgical strategy should be planned in advance. When managing a patient with an unanticipated PAS for delivery, there are four surgical options. Ideally, a surgeon with specific surgical PAS experience should perform the hysterectomy. A second option is pre-delivery intra-operative transfer. The abdomen would be closed prior to cesarean delivery when PAS is suspected and the woman transported to a higher-level center. A third option in a hemodynamically stable patient is post-delivery, pre-hysterectomy transfer to a higher-level center for

planned hysterectomy on arrival there. A fourth option is conservative expectant management, in which the undisturbed placenta is left in situ after the birth of the newborn, with planned prolonged follow-up at a specialist center. In more dire circumstances, such as cases of DIC following hysterectomy and a limited blood bank inventory, abdominal packing followed by urgent transport to a higher-level center is advised.

Hemorrhage management and monitoring

Prevention and management of hemorrhage

Due to lack of robust evidence, recommendations for hemorrhage management come primarily from expert opinion and consensus statements.^{66,67} Close communication with the surgical team is vital to obtain regular updates on the extent, source, and surgical control of bleeding sites. Approaches to the intra-operative management of severe hemorrhage are discussed in another Special Issue review and here we highlight key aspects of hemorrhage control specific to women with PAS.

Surgical hemorrhage control

In cases of massive pelvic hemorrhage and uncontrollable DIC, abdominal and pelvic packing may be necessary.^{68–71} After hysterectomy and securing all arterial vasculature, the surgical team packs, covers, and seals the open abdomen, and applies negative pressure to the dressing via intermittent wall suction to maintain pressure on the bleeding surfaces of the pelvis. This temporizing measure enables constant tamponade while blood products and clotting factors are replaced in the intensive care unit.

Vascular access

Establishing wide bore intravenous access before surgery is essential. In a separate survey of Israeli lead anesthesiologists,⁵⁷ 83% of respondents placed at least two peripheral intravenous lines for low suspicion PAS cases and 15% placed a central vein line for high suspicion cases. Invasive lines placed at our institutions are summarized in Table 3. A peripheral rapid infuser catheter (7 or 8.5 French Gauge) placed under ultrasound guidance should be considered for women with difficult intravenous access. The authors rarely place central venous lines, however accept that this practice may be different at other institutions.⁶⁰

Access to a cell salvage and a rapid infuser device, e.g. Level 1 (Smiths-Medical, Minneapolis, Minnesota, USA) or Belmont (Belmont Medical Technologies, Berkshire, United Kingdom), is also recommended.

Monitoring

Direct invasive arterial monitoring enables rapid assessment of hemodynamic changes and the ability to draw blood samples for laboratory testing. Based on data from an Israeli survey⁵⁷ and a case series by Taylor et al.⁶⁰ more than 75% of PAS cases received invasive arterial monitoring. Additional monitoring may include non- or minimally-invasive cardiac output monitors and transthoracic or transesophageal echocardiography. The bedside Rapid Obstetric Screening Echocardiography scan, systematically assessing contractility and volume status, and a right heart screen for embolism, has been described.⁷²

Communication with blood bank

Before or during a major obstetric hemorrhage, close communication with blood bank personnel is vital. In preparation for major

hemorrhage, blood products should be ordered pre-operatively.⁶⁶ For women who undergo scheduled cesarean hysterectomy, type-specific or crossmatched blood products can be ordered in advance, with typical orders for a scheduled cesarean hysterectomy at our institutions described in Table 3. Other centers may order a larger number (e.g. 12) packed red blood cells (PRBCs) before surgery and use a temperature-controlled cold-storage box (Igloos) to reduce wastage of unused products.⁶⁶

Critical to blood management during urgent or emergency cesarean hysterectomy is timely access to a sufficient volume and various types of blood products. An institution-specific massive transfusion protocol (MTP) ensures a short turn-around time between the ordering and receipt of emergency-release blood products. Blood products provided by the MTP likely vary between institutions. For example, at Stanford University Medical Center (California, US), the MTP contains six units of un-crossmatched or type-specific PRBCs, four units of liquid fresh frozen plasma, and one apheresis platelet unit. Standardized ordering is an essential component of a MTP that limits delays and avoids miscommunication between care teams in the operating room and personnel in the blood bank. Lastly, a MTP is a critical resource when crossmatched blood products have been exhausted during major obstetric hemorrhage. It allows the care team to keep up with ongoing demands for blood products until surgical and hemostatic control of bleeding has been achieved.

Point-of-care devices such as thromboelastometry and arterial-blood gas machines can be used to guide blood product management. Information about the maternal coagulation profile will guide transfusion decision-making and reduce blood product requirements, with associated cost savings.^{66,73,74} Targeted (or goal-directed) blood product administration based on viscoelastic and laboratory data has been described for obstetric hemorrhage management.⁷⁵ Other reviews in this Special Issue discuss and compare the utility of a fixed ratio approach with a goal-directed approach and the management of hemorrhage-related coagulopathy. Other important considerations are the close monitoring of ionized calcium and magnesium, and aggressive treatment of hypocalcemia⁷⁶ and hypomagnesemia during massive transfusion.

Antifibrinolytics such as tranexamic acid have not been studied in PAS surgery. By extrapolating data showing clinical benefit and lack of harm in trauma surgery and postpartum hemorrhage,⁷⁷ it is reasonable to consider using antifibrinolytics during hemorrhage in PAS cases. In other settings the risk of death from exsanguination is reduced if tranexamic acid is administered within 3 h of bleeding onset.⁷⁸ Data on the prophylactic administration of tranexamic acid are lacking.

The efficacy of uterotonics such as oxytocin and locally-acting hemostatic agents has not been investigated in PAS cases. The potential benefit of improving uterine contractility with oxytocin appears offset by the potential for placental detachment. In our practice, prophylactic oxytocin regimens are used routinely after delivery of the neonate.

Postoperative care

Close patient monitoring is recommended for the first 72 h following delivery for several reasons. Potential postoperative complications requiring medical or surgical intervention include: ongoing bleeding, coagulopathy, re-operation, respiratory failure from transfusion complications including transfusion-related acute lung injury, transfusion-associated circulatory overload, surgical site infection, bowel injury, urinary tract injury, and vesico-vaginal fistula. Bladder injury and repair occurs in 15–30% of PAS surgeries, so prolonged indwelling urinary catheterization may be required to rest the bladder. Uterine artery embolization may be required to control intra-abdominal bleeding in hemodynamically stable patients. Postoperative thromboprophylaxis with prophylactic low molecular weight heparin regimens should be considered until patients are able to ambulate independently. Based on guidelines from the Society of Obstetric Anes-

thesia and Perinatology, the first postoperative dose can be given at least 12 h after neuraxial blockade or at least 4 h after epidural catheter removal following hysterectomy, in the absence of ongoing or active bleeding.⁷⁹ In Utah, prophylactic enoxaparin is initiated 6–12 h post-surgery (with the above caveats) providing bleeding concerns have passed and the patient is hemodynamically stable. Enoxaparin is continued for 2–4 weeks postoperatively, depending on individual considerations such as blood loss, emergent surgery, personal history, and the extent of ambulatory effort.

Postoperative analgesia regimens may vary based on factors such as the type of surgical incision (lower and/or upper midline incision), use of long-acting neuraxial opioids, and the presence of an epidural catheter. Options include neuraxial morphine, epidural analgesia (infusion and/or patient controlled epidural analgesia), truncal blocks (such as transversus abdominis plane block or quadratus lumborum block), and patient-controlled intravenous opioid analgesia. Those used at our institutions are shown in Table 3.

In our experience, patients are often discharged 3–5 days after surgery, after meeting standard milestones (ambulating independently, pain controlled on oral medications, voiding if not on bladder rest, passing flatus). Following a traumatic delivery, women frequently experience postpartum psychopathologies.^{80,81} As recommended by the ACOG, the 6-week postpartum assessment should include a screening tool for postpartum depression.⁸² Women should also be screened for other postpartum psychopathologies such as post-traumatic stress disorder or anxiety.

Box of recommendations

- Patients with any clinical risk factors for PAS should be referred to a specialist center for formal diagnosis and multidisciplinary care management. These centers should comprise an experienced multidisciplinary team, enhanced diagnostic and surgical resources, and a blood bank which can support the need for massive transfusion.
- Surgical and anesthesia teams may need to be mobilized at short notice in the event of urgent or emergency cesarean hysterectomy.
- Every center with a labor and delivery unit should have a protocol, including pathways for referral and transfer, for women with PAS who require urgent or emergency delivery.
- Antenatal diagnosis is currently hampered by underdiagnosis or misdiagnosis.
- Decision-making related to the timing of delivery should account for the risks of prematurity from a scheduled delivery before 35 weeks' gestation versus major hemorrhage if urgent delivery is required after 35 weeks' gestation.
- Delivery in an operating room in the labor and delivery unit or the main operating room complex, or a hybrid operating room (main operating room or interventional radiology unit). The decision should be based on local resources and providers' preferences.
- Consensus is lacking on the optimal anesthesia mode for women with PAS requiring cesarean hysterectomy. Options include: neuraxial anesthesia (with a catheter-based technique), general anesthesia, or combined neuraxial and general anesthesia.
- After surgery, care teams should ensure that patients are closely monitored for major surgical complications, including bleeding and surgical site infection.
- Postoperative analgesia options include: neuraxial morphine, epidural analgesia (patient-controlled analgesia and/or an infusion), truncal blocks, and regular acetaminophen and non-steroidal anti-inflammatory agents. Multimodal analgesia regimens are recommended.

Conclusion

Placenta accreta spectrum is the leading cause of peripartum hysterectomy in developed countries, and an increasingly important con-

tributor to maternal death and maternal morbidity worldwide. To optimize peri- and postpartum outcomes of women with PAS, obstetric anesthesiologists should be familiar with all aspects of peri-operative planning and management, particularly anesthesia mode decision-making and the management of massive hemorrhage (see Box of Recommendations).

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